

# Predicting Customer Churn Using Classification Algorithms

Case: IBM Telco Customer Churn Dataset | Sample: 7,043 Customer Records

Methods: Logistic Regression · Decision Tree Classification · Confusion Matrix · ROC-AUC  
Evaluation

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## 1. Objective

The primary objective of this study is to predict whether an individual telecom customer will discontinue their subscription (churn) using demographic, account, and service-usage attributes from the IBM Telco Customer Churn dataset. Two classification frameworks are built, trained, and compared: a Logistic Regression model, which provides a linear, probabilistic baseline, and a Decision Tree Classifier, which captures non-linear, rule-based interactions between predictors. The study evaluates both models on a common test set using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and the Confusion Matrix, and recommends the model best suited for operational churn-prediction and retention targeting.

## 2. Data and Method

The dataset comprises 7,043 individual customer records with 21 attributes, and no missing values or duplicate rows were found during data quality checks. The variables span four broad categories:

- Target Variable: Churn (Yes/No), recoded as a binary numeric variable (1 = Churned, 0 = Retained) for modelling.
- Demographic Variables: Gender, Senior Citizen status, Partner, and Dependents.
- Account Variables: Tenure (months), Contract type, Paperless Billing, Payment Method, Monthly Charges, and Total Charges.
- Service Usage Variables: Phone Service, Multiple Lines, Internet Service type, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies.

Of the 7,043 customers, 5,174 (73.5%) were retained and 1,869 (26.5%) churned, indicating a moderate class imbalance that is accounted for when interpreting Recall and F1-Score.

### 2.1 Data Preprocessing

- The non-informative customerID column was dropped prior to modelling.
- The target variable Churn was mapped from Yes/No to 1/0.
- All categorical predictors were converted into dummy variables using one-hot encoding (drop\_first = True) to avoid the dummy-variable trap.
- The data was split into training (80%) and testing (20%) sets using a fixed random\_state = 42 for reproducibility, producing 5,634 training records and 1,409 test records after one-hot encoding of categorical variables.

### 2.2 Modelling Approach

Each research question requires a classification approach suited to the nature of the decision boundary being modelled:

- Logistic Regression is applied as a baseline linear classifier that models the log-odds of churn as a linear combination of predictors, offering interpretable coefficients and probability estimates.
- Decision Tree Classification is applied to capture non-linear interactions and threshold effects among predictors (e.g., tenure and contract type), and to surface feature importance rankings.

Both models are trained on the identical training set and evaluated on the identical held-out test set, ensuring a fair, like-for-like comparison of predictive performance.

### 3. Results

#### 3.1 Exploratory Data Analysis (EDA)

A correlation heatmap of the numeric variables (Senior Citizen, Tenure, and Monthly Charges) shows only weak-to-moderate pairwise correlations (all below 0.25), indicating limited multicollinearity among the raw numeric predictors.

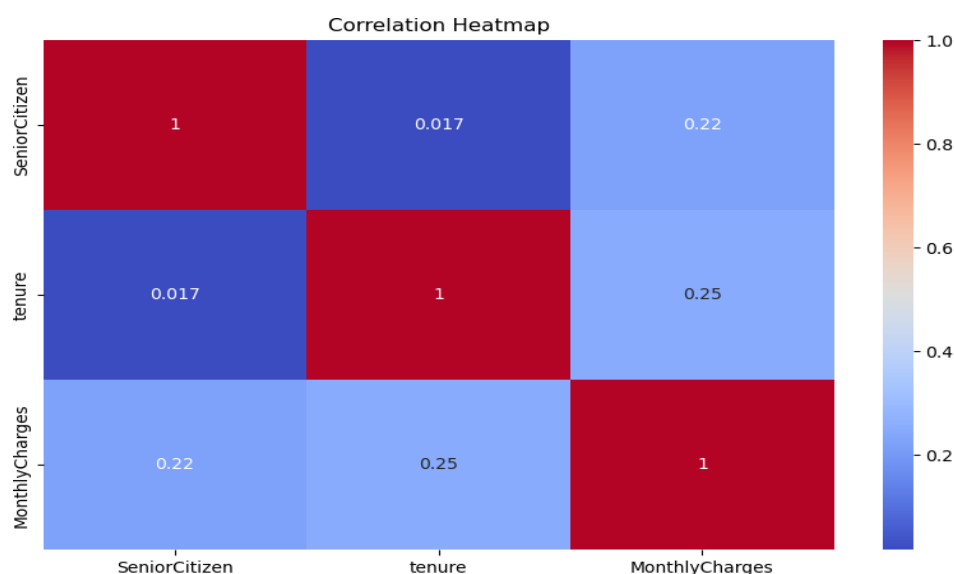


Figure 1: Correlation Heatmap of Numeric Variables

Histograms of the numeric variables reveal that Tenure and Monthly Charges are broadly dispersed rather than normally distributed, with a notable spike of customers at very low tenure, consistent with a segment of recent sign-ups that are more churn-prone.

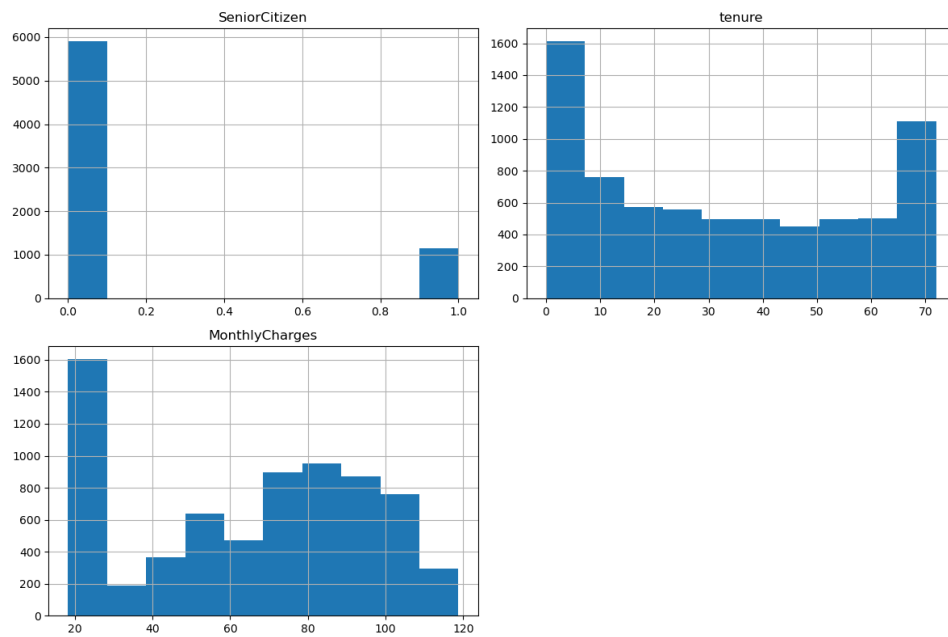


Figure 2: Distribution of Numeric Variables

The target variable is moderately imbalanced: substantially more customers are retained than churned, which is factored into the evaluation by emphasising Precision, Recall, and F1-Score alongside Accuracy.

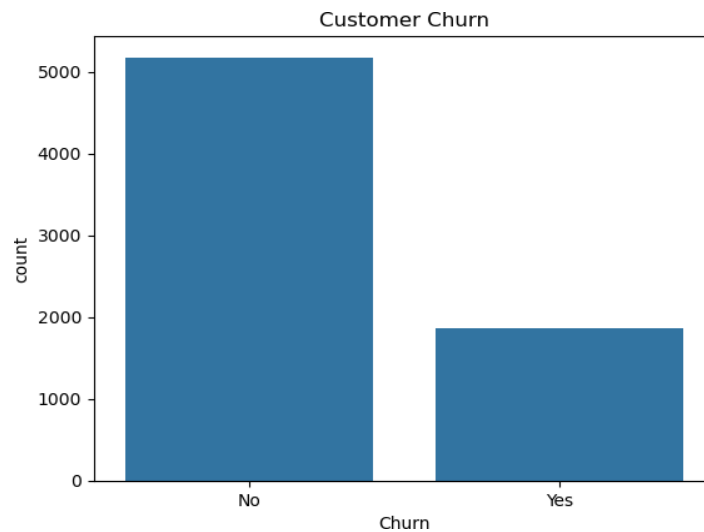


Figure 3: Class Distribution of Churn

A scatter plot of Tenure versus Monthly Charges, coloured by churn status, shows that churned customers cluster at low tenure across a wide range of monthly charges, while long-tenure customers churn far less frequently regardless of their charge level.

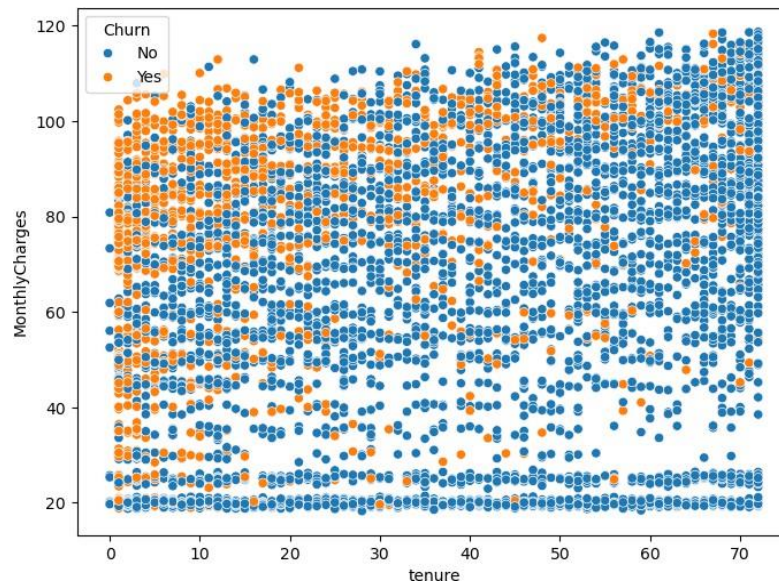


Figure 4: Tenure vs. Monthly Charges by Churn Status

A boxplot comparison confirms that churned customers tend to pay higher Monthly Charges on average, and exhibit a wider interquartile spread than retained customers.

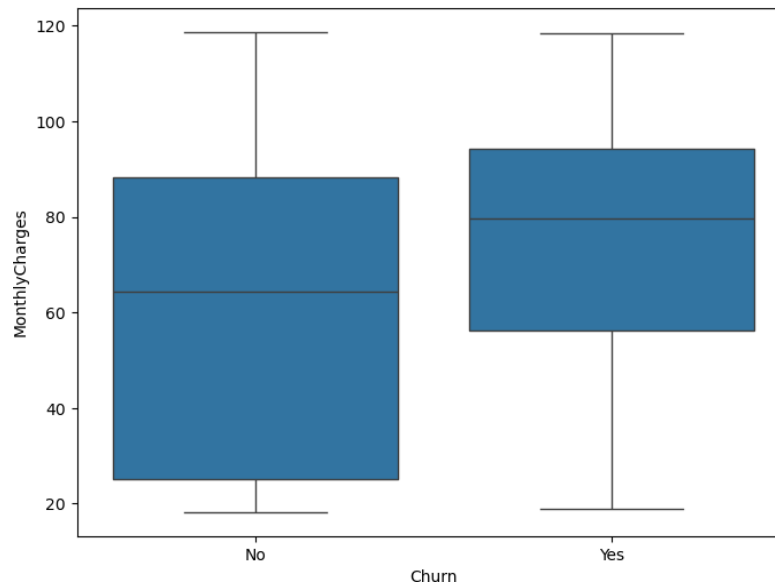


Figure 5: Monthly Charges by Churn Status

### 3.2 Model 1 — Logistic Regression

$H_0$ : The model provides no better classification of churn than chance.

$H_1$ : The model provides significantly better classification of churn than chance.

Performance Metrics

| Metric    | Value  |
|-----------|--------|
| Accuracy  | 0.8105 |
| Precision | 0.6596 |
| Recall    | 0.5871 |
| F1-Score  | 0.6213 |
| ROC-AUC   | 0.7390 |

Table 1: Logistic Regression — Test Set Performance Metrics

Confusion Matrix

|                  | Predicted: No Churn | Predicted: Churn |
|------------------|---------------------|------------------|
| Actual: No Churn | 923                 | 113              |
| Actual: Churn    | 154                 | 219              |

Table 2: Logistic Regression — Confusion Matrix

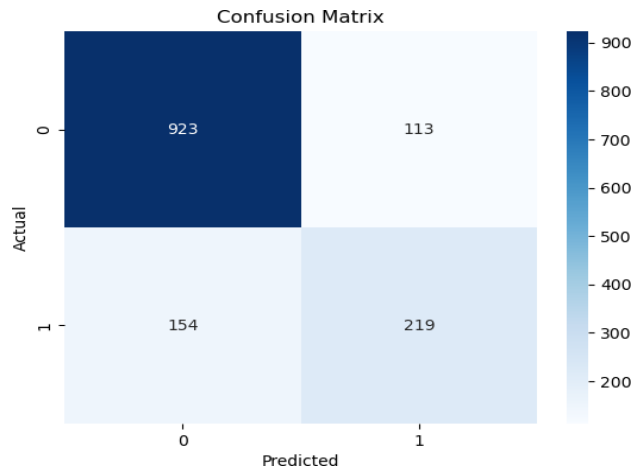


Figure 6: Logistic Regression Confusion Matrix Heatmap

Interpretation: The Logistic Regression model correctly classifies 81.1% of test customers overall. Of the customers it predicts will churn, 66.0% actually do (Precision), and it successfully identifies 58.7% of all customers who actually churned (Recall). The ROC-AUC of 0.739 indicates the model discriminates meaningfully better than random guessing (AUC = 0.5) between churners and non-churners.

3.3 Model 2 — Decision Tree Classification

Performance Metrics

| Metric    | Value  |
|-----------|--------|
| Accuracy  | 0.7729 |
| Precision | 0.5875 |

| Metric   | Value  |
|----------|--------|
| Recall   | 0.4772 |
| F1-Score | 0.5266 |
| ROC-AUC  | 0.6783 |

Table 3: Decision Tree — Test Set Performance Metrics

Confusion Matrix

|                  | Predicted: No Churn | Predicted: Churn |
|------------------|---------------------|------------------|
| Actual: No Churn | 911                 | 125              |
| Actual: Churn    | 195                 | 178              |

Table 4: Decision Tree — Confusion Matrix

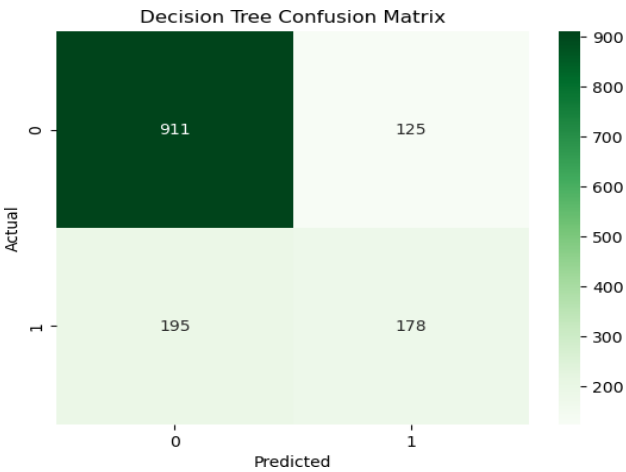


Figure 7: Decision Tree Confusion Matrix Heatmap

Interpretation: The unconstrained Decision Tree achieves 77.3% overall accuracy — lower than Logistic Regression on every metric. Its Recall of 47.7% is notably weaker, meaning it misses a larger share of customers who actually churn, which is a material limitation for a retention-targeting use case.

Feature Importance (Top 10 Predictors)

| Rank | Feature                          | Importance |
|------|----------------------------------|------------|
| 1    | Tenure                           | 0.2172     |
| 2    | Monthly Charges                  | 0.1249     |
| 3    | Internet Service: Fiber Optic    | 0.1038     |
| 4    | Payment Method: Electronic Check | 0.0167     |
| 5    | Paperless Billing: Yes           | 0.0154     |
| 6    | Senior Citizen                   | 0.0153     |

| Rank | Feature            | Importance |
|------|--------------------|------------|
| 7    | Partner: Yes       | 0.0150     |
| 8    | Contract: One Year | 0.0109     |
| 9    | Tech Support: Yes  | 0.0092     |
| 10   | Gender: Male       | 0.0089     |

Table 5: Decision Tree — Top 10 Feature Importances

Interpretation: Tenure and Monthly Charges are, by a wide margin, the two most influential drivers of the tree's churn predictions, together accounting for over a third of total feature importance. Subscribing to Fiber Optic internet is the third-strongest signal, followed by a long tail of smaller but statistically relevant contract, billing, and demographic factors.

3.4 Model Comparison — Logistic Regression vs. Decision Tree

| Metric    | Logistic Regression | Decision Tree |
|-----------|---------------------|---------------|
| Accuracy  | 0.8105              | 0.7729        |
| Precision | 0.6596              | 0.5875        |
| Recall    | 0.5871              | 0.4772        |
| F1-Score  | 0.6213              | 0.5266        |
| ROC-AUC   | 0.7390              | 0.6783        |

Table 6: Side-by-Side Model Performance Comparison

Logistic Regression outperforms the Decision Tree Classifier on every single evaluation metric — Accuracy is higher by 3.8 percentage points, Recall by 11.0 points, F1-Score by 9.5 points, and ROC-AUC by 6.1 points.

3.5 ROC Curve — Logistic Regression

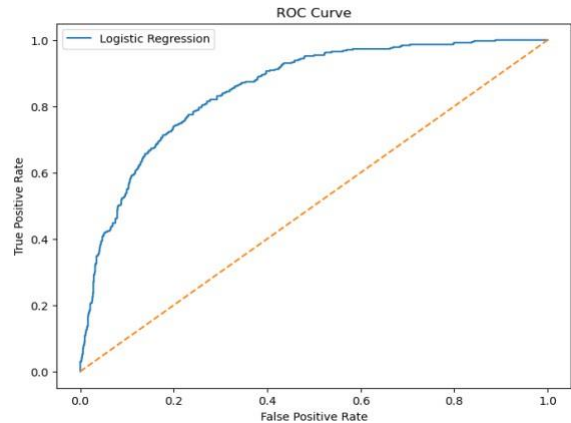


Figure 8: ROC Curve for the Logistic Regression Model

The ROC curve for the Logistic Regression model sits comfortably above the diagonal chance line across the full range of decision thresholds, visually confirming the model's ROC-AUC of 0.739 and its ability to separate churners from non-churners better than a random classifier.

### 3.6 Threshold Sensitivity Analysis (Logistic Regression)

Because the default 0.5 probability cut-off is not necessarily optimal for a business that wants to catch as many at-risk customers as possible, three alternative classification thresholds were evaluated:

| Threshold | Accuracy | Precision | Recall | F1-Score |
|-----------|----------|-----------|--------|----------|
| 0.3       | 0.7580   | 0.5285    | 0.7962 | 0.6353   |
| 0.5       | 0.8105   | 0.6596    | 0.5871 | 0.6213   |
| 0.7       | 0.7729   | 0.7732    | 0.2011 | 0.3191   |

Table 7: Classification Threshold Comparison

Interpretation: Lowering the threshold to 0.3 substantially raises Recall to 79.6% (catching far more true churners) at the cost of Precision, which falls to 52.9% meaning more false alarms among the customers flagged for retention outreach. Raising the threshold to 0.7 achieves the highest Precision (77.3%) but Recall collapses to just 20.1%, missing the vast majority of customers who actually churn. The default 0.5 threshold offers the best-balanced F1-Score (0.6213) of the three, but the 0.3 threshold may be preferable in practice if the cost of a missed churner outweighs the cost of an unnecessary retention offer.

## 4. Discussion

### 1. Which model performs best?

Logistic Regression consistently outperforms the Decision Tree Classifier across Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The unconstrained Decision Tree likely overfits the one-hot-encoded feature space, reducing its ability to generalise effectively to the held-out test data. In contrast, the more regularised Logistic Regression model demonstrates stronger and more consistent predictive performance across all evaluation metrics, making it the more reliable model for customer churn prediction.

### 2. What are the key drivers of customer churn?

Tenure and Monthly Charges are the dominant predictors of churn: customers with short tenure and higher monthly charges are markedly more likely to leave. Subscribing to Fiber Optic internet service is the third-most important signal, plausibly reflecting service-quality or pricing dissatisfaction among fibre customers. Contract type, payment method (particularly electronic check), paperless billing, and senior citizen status also contribute meaningfully to the churn decision.

### 3. Does the choice of classification threshold matter for business decisions?

Yes. The threshold sensitivity analysis shows a clear Precision-Recall trade-off: a lower threshold (0.3) is better suited to a proactive retention campaign where missing a genuine churner is costly, while a higher threshold (0.7) suits a resource-constrained team that can only act on its highest-confidence predictions. The choice should be driven by the relative cost of a false negative (a lost customer) versus a false positive (an unnecessary retention incentive).



#### 4. Which model would you recommend, and why?

We recommend deploying the Logistic Regression model as the primary churn-classification tool, for the following reasons:

- Superior and more balanced performance — it leads on every metric, including the business-critical Recall and F1-Score.
- Interpretability — its coefficients and predicted probabilities can be directly translated into a churn-risk score that supports transparent, explainable retention decisions.
- Threshold flexibility — its continuous probability output allows the business to tune the classification threshold to match its specific retention-cost trade-offs, as shown in Section 3.6.

The Decision Tree remains valuable as a diagnostic tool, since its feature-importance ranking offers an intuitive, rule-based view of which customer attributes matter most — insight that can guide targeted retention strategy even where the tree itself is not the deployed classifier.

#### Conclusion

The empirical analysis demonstrates that classification modelling can effectively predict customer churn using demographic, account, and service-usage attributes from the Telco dataset. Logistic Regression provides a superior, well-balanced, and interpretable classifier (Accuracy = 0.8105, ROC-AUC = 0.7390) relative to the Decision Tree (Accuracy = 0.7729, ROC-AUC = 0.6783), while the tree's feature-importance output confirms that tenure, monthly charges, and fibre-optic internet subscription are the leading drivers of attrition. Incorporating threshold tuning alongside the base model further allows the business to align churn-prediction outputs with its specific cost trade-offs between missed churners and unnecessary retention spend, delivering a practical framework for proactive customer-retention strategy.